Interim verification and validation review · V8.3

Taif / Palm House · 3 Nov 2026 · review only, no design file changed · inputs: `docs/REQUIREMENTS.md` (R-01 … R-37), the plans, the studies, `data/decisions.csv`, `docs/OPEN_ITEMS.md`, the V6 review (`docs/review/design_review.typ`, tag `review-1`) and its status after V8.3 · written for the owner to read on his return

Executive summary

Overall verdict. V8.3 is a coherent, well-documented concept design whose **house works**: the guest / family separation, the courtyard heart, the master suite, the maid wing, the fire strategy at home level and the two plumbing clusters all walk through their scenarios. It is **not ready to freeze for V9.0**, for four reasons that have nothing to do with drawing quality: (1) the plot's setback margin is still provisional (V-01, open since V6); (2) nothing structural has been checked by an engineer and there is no soil report (V-02); (3) the owner's 3 Nov decisions left **bedroom 1 and the future kids' bedroom without a shower on their floor** (V-03), a validation failure for this family; (4) the services concept has **no heating and no energy-code envelope** (V-04) and **no water in a power cut** (V-05). The remaining majors are site and yard conflicts created by the latest moves (the parking bay against the north windows V-07, the hood exhaust beside the outdoor counter V-08) and two accepted residual risks the owner should see in writing (the open stair V-09, the undecided suite-bath drainage and the overfull utility room V-10). Of the 37 requirements, 24 are met, 9 partly, 1 not met (R-11), 3 not verifiable yet. Findings: 2 critical · 8 major · 8 minor · 2 suggestions.

The top 10 issues by severity

No.	Sev.	Where	Issue	Recommended fix
1	critical	Site · the setback margin	**V-01** · The nearest outer wall face stands 0.20 m from the lane-side setback line (0.80 m from the north street line), a **provisional** margin set in V6.3 without the official survey. A survey difference of a few centimetres puts the wall in the setback; the whole plan is set out from peg M1 and the envelope of `base.typ`, neither confirmed. Three versions later this is still open (V6 review S-03, critical).	Commission the legal survey before any further geometry work; if the margin is wrong, move the house south (the garden has ≈ 7 m) or trim the maid wing. Effect: everything downstream (the DXFs, the 3D, the site plan) is parametric on the shift, so a move is cheap now and very costly after the engineers start.
2	critical	Structure · unverified concept, no soil report	**V-02** · Every structural size is assumed: the column grid, the 1.55 m west projection on an edge beam and three columns, the ribbed slabs, the men's majlis roof carrying the gym, the suite court terrace and the office (X5, "revised, improved, **not verified**"), the stair trimmers, the new B14 condition (X21). No soil report exists; the pads, the lift pit, the breeze channel in the slab and the soak-aways are drawn on an assumed bearing stratum. The design has progressed to interiors and plumbing on a structure nobody has checked.	Appoint the structural engineer now with the soil investigation; freeze nothing that depends on beam depths (the ceiling heights under B4 / B5 / B6, the 2.14 m stair headroom, the 0.15 depressed suite bath zone) until the first sizing round. Effect: possible deeper beams under the first-floor west rooms and the suite walls; the stair trimmer option 0.30.
3	major	FF · bedroom 1 (and the kids' room) without a shower	**V-03** · V8.3 turned the family bath into a WC and gave bedroom 2 the shower. Bedroom 1 (and the kids' room when it becomes a bedroom) now has **no shower on its floor**: its users must enter bedroom 2's or bedroom 3's en-suite or the suite. The hotel-layout en-suite B4 cannot fit the wedge room (its study page 3); the ground floor has no shower by decision. The house therefore has a bedroom whose daily route to a shower crosses another child's bedroom: a validation failure for a 5–6 person family with children of mixed ages (R-01, R-09).	Decide now, before V9.0: (a) B4-b for bedroom 1 with its 7.3 m² L bedroom (the owner accepts the two failed criteria), (b) a shower back in the family WC by taking 0.6 m from bedroom 2's bath (bedroom 2's shower 1.0 × 1.0: tight), (c) bedroom 1 declared the guest / older-child room that shares bedroom 3's en-suite through a second door (bedroom 3's bath door is at x 2.2 inside bedroom 3's entry, far from bedroom 1), or (d) accept the walk. Effect: (a) bedroom 1's furniture and the laundry's corner drop; (b) bedroom 2's bath; (c) none.
4	major	Services · no heating, no energy-code envelope	**V-04** · The services concept covers ventilation (HRV), cooling (AC condensers on the roof) and the courtyard breeze; **heating is absent**. Taif at ≈ 1,700 m has winter nights near 0–5 °C; a courtyard house with large north-facing courtyard glass (W1 / W2 / W3, single steel-framed divided lites with no thermal specification) and an open-sky courtyard needs a heating strategy (reversible heat pumps, underfloor heating in the living and the baths) and an envelope specification (insulation, glazing U-value, thermal bridges at the steel windows, airtightness). None exists.	Add a heating concept to the MEP brief (reversible VRF or split heat pumps; underfloor heating in the living, the courtyard-facing rooms and the bathrooms), specify thermally broken steel windows and the wall / roof insulation before the facade stage; check the WWR of the courtyard glass against SBC 602. Effect: the roof plant zone (heat pumps), the slab build-up (underfloor heating thickness: the finish 0.08 assumed), the window frames' depth.
5	major	Services · water and power in a power cut	**V-05** · The owner removed the generator (V8.2); the PV batteries are a later installation. Until then a power cut stops the booster pump (the only water supply is the ground tank 38 m away through a pump), the HRV, the lift and the assist fan. The roof study reserves a **water tank zone** between R10 and R3, but the plumbing study's water scheme (tank → booster pump → manifolds) has **no roof tank**, so there is no gravity water in a cut and the "water from the roof tank" scenario cannot be walked. The lift's battery lowering, the alarm panel's battery and the gas valve's fail-safe are stated; the water is not.	Decide the water system now: a roof tank (≈ 2 m³, the zone exists) fed by the booster, gravity to the fixtures with a small pressure set only for the showers; an essentials circuit sized for the batteries (pumps, fridges, the HRV, lights, the lift lowering). Effect: the roof structure (R10 / R3 take ≈ 25 kN), the attic's hiding height (1.40 m: a 2 m³ tank is ≈ 1.2 m high: check), the plumbing risers in the west shaft.
6	major	Site · the septic and the long foul main	**V-06** · The existing septic lies **outside the farm wall near corner C1**, under the street side: its legal status for a new house, its inlet level, its capacity and its condition are unknown. The east main runs 36 m north and west with its inlet computed at −1.19 below the house floor on assumed falls; if the surveyed inlet is higher, a sewage pump station at the gate is needed, with all its maintenance and power dependence (V-05). The foul drains also cross under the boundary wall and run in the driveway under the guest path.	Survey the septic first (position, inlet level, size); ask the municipality; if a pump is needed, consider a new septic / treatment unit inside the farm south of the house where the levels work by gravity. Effect: the whole foul route, the gate chamber, the driveway works.
7	major	Site · the north parking bay against the north facade	**V-07** · The kept family car bay stands 0.4 m from the house's north wall, directly in front of the women's majlis north window (x 5.4–7.8) and under the kids' room's north window. A parked car 0.4 m from a window gives the room a bumper view, exhaust and heat,	Move the bay east along the strip (x 10–15, in front of the maid wing's cut wall, where the windows are small and obscured) or drop it and keep the farm parking; keep a 1.5 m planted strip at the majlis window. Effect: the swept

			and it contradicts the new facade principle (R-37) that wants the north facade **opened** for soft north light. The site plan also makes the north strip "a calm family space" and parks a car in it.	path (the strip is 4–4.6 m deep: a bay further east needs the same 3-point manoeuvre), the workers' route along the boundary wall.
8	**major**	East yard · the hood exhaust beside the outdoor counter; the bins at the maid's window	**V-08** · The outdoor counter and sink now sit under the kitchen window at y 8.3–10.1; the hood's exhaust outlet and the make-up air inlet are on the same wall at y 10.2–11.2, 0.1 m from the counter's south end. Cooking fumes and grease exhaust blow across the outdoor kitchen, and the make-up inlet draws the outdoor kitchen's smoke into the house. The bins (y 5.3–6.1) stand ≈ 1 m from the maid room's escape window on the cut wall (y 3.9–4.9 CSV).	Either move the counter north of the service door (y 5.8–7.0, swapping with the bins: the bins then go south, screened) or raise the hood outlet to the band (+3.4) with a vertical discharge and move the make-up inlet to the cut wall's north part; move the bins ≥ 3 m from the maid's window. Effect: the yard plan, the hood duct (a 2 m rise inside the kitchen's east wall zone), the gas pipe route.
9	**major**	Fire · one open stair in the riwaq, smoke from the kitchen	**V-09** · The only stair is open to the riwaq (no enclosure, no fire door, by the owner's home-level decision). A night fire in the kitchen reaches the dining and the double-height living (one room, no door between the living and the dining), the riwaq through the living's arched entrance (no door) and the stair well in minutes; the first floor's alternative exit is the escape window of each sleeping room (a 1.6 m drop from the sill to the ground is the drop from a first-floor window: ≈ 4.5 m to the ground). The clerestories vent the living's smoke only if someone opens them. This is the owner's accepted residual risk; the review records that the strategy relies entirely on early detection and on the windows.	Keep the decision but harden it: a heat alarm in the kitchen (drawn), a **closer on the kitchen's dining door** and a door at the living's arch is the cheapest protection of the riwaq and the stair; an escape ladder at bedroom 1's and the kids' room's windows; the clerestories on an automatic smoke-vent actuator linked to the alarms. Effect: the arch gets a glazed door (the design language accepts black steel); the clerestory actuators (open item 14 closed as openable: add the actuator).
10	**major**	Suite bath over the kitchen · the drainage approach undecided; the utility room overfull	**V-10** · The one wet room over a habitable room is the suite bath over the kitchen. Two approaches are drawn ((a) every drain in the 0.15 depressed slab, the WC and the tub at the limit of the 0.14 m pipe zone; (b) an east service wall with the tub on a plinth, one local opening left); the owner has not decided, and the kitchen's ceiling, the hood zone and the east stack's branches depend on it. The utility room (2.5 × 2.5 axes) must hold the shaft, the main panel, the inverter and battery cabinet, the heater, the manifold cabinet, the filter, the booster connection, the alarm panel and the recorder: ≈ 4.3 m² of free floor, flagged small by the plumbing study itself.	Decide (b) before V9.0 (the review agrees with the study: accessible, nothing in the kitchen ceiling); move the batteries to the store under the stair's half landing with a fire-rated door, or enlarge the utility room by 0.5 m into the hospitality kitchen. Effect: the hospitality kitchen's counter (2.5 → 2.0 m), the stair store.

Top strengths

- **Separation that works without a word:** one entry, the guest foyer with its majaz, the men's majlis entered only from the foyer, the two-door serving passage, the solid TV wall; no guest sees a family room, the courtyard or the family's cars (R-14, R-15: met and validated in the weekly-guests scenario).
- **The courtyard as a room:** 28 m² level, enclosed by W1 / W2 / W3 so the house is closed and filtered round an open sky (V6's critical Y-01 is resolved), the breeze path and the TV wall, now with movable seating and the water and tree in the kitchen's view.
- **Plumbing discipline:** two clusters, no third; every first-floor wet room over a wet or service room except the suite bath; the first-floor family WC and bedroom 2's bath on one service wall into the shaft; rain and sewage separated; the smells list.
- **Documentation and traceability:** every decision dated and reasoned, every study with its own checks, the plans, the DXFs and the 3D generated from one data set (the consistency section found only record-keeping mismatches, no geometric ones).
- **Fire at home level, honestly stated:** interconnected alarms, an escape casement in every sleeping room, the suite court as a refuge, the roof ladder not an exit.

الخلاصة التنفيذية: تصميم V8.3 متماسك وموثق ويعمل البيت فيه كما طُلب: فصل الضيوف عن العائلة، الفناء قلب البيت، الجناح، جناح الخادمة، السلامة من الحريق على مستوى المنزل، وعنقودا السباكة. لكنه غير جاهز للتجميد في V9.0 لأربعة أسباب: هامش الارتداد ما زال مؤقتا بلا رفع مساحي رسمي؛ لا يوجد تدقيق إنشائي ولا تقرير تربة؛ قرارات 3 نوفمبر تركت غرفة النوم 1 (وغرفة الأطفال مستقبلا) بلا دش في دورها؛ ولا يوجد تدفئة ولا مواصفات غلاف حراري، ولا ماء عند انقطاع الكهرباء. أهم عشر مسائل في الجدول أعلاه، والنتائج كاملة في القسم 7.

1 · Verification · requirements traceability matrix

Every requirement of `docs/REQUIREMENTS.md` → where it is met → status → evidence. "Met" means the files show it as drawn or decided, not that a code authority has accepted it.

Req.	Requirement (short)	Where it is met	Status	Evidence / remark
R-01	5–6 residents, children share	FF: suite + bedrooms 1–3, the kids' room a 4th bedroom later	met	Bedroom 3 14.97 m² clear takes two beds (`tools/bedroom_test.py`); bedroom 2 12.84; bedroom 1 13.83; kids' 12.44. **But** see V-03: not every bedroom has a shower on its floor.
R-02	Maid wing ≥ 6.5 m², ≥ 2.1 m, own bath, private route	GF maid room + bath, `east_wing.typ`	met	7.28 m² clear, 2.49 m at the bed (V7.0); own bath 3.03 (door 0.70: V-11); the route gate → yard → service door → kitchen → laundry never crosses the family.
R-03	No driver room	—	met	None drawn.
R-04	No GF bedroom, no prayer room	—	met	None drawn.
R-05	Medium storage	Linen 0.95, the coat cupboard 1.26 × 1.45, the store under the half landing 2.5 × 1.25, pantry 1.8 m	partly	No outdoor / garden store and no general store ≥ 4 m² (the under-stair store is 1.65 m high); the proxy's "general store" is not met.
R-06	Lift at construction	GF / FF lift 2.25 × 2.2 placeholder, `design_gf.typ`	partly	Decided V6.3; the shaft awaits the supplier (V-20); car ≥ 1.1 × 1.4 plausible, not verified.
R-07	Doors ≥ 0.85, corridors ≥ 1.2	Every bedroom / bath door 1.00 opening; the gallery 1.405 (1.235 at columns); the riwaq ≈ 1.3; the serving passage 1.2; the family link 1.0	partly	Exceptions: the guest WC 0.90 opening, the maid bath 0.70 (V-11); the link and the gallery at the columns at the 1.0 minimum (V-12); the 1.5 m turning circle in the powder room ✓ and at the lift ✓.
R-08	Step-free routes	GF level; the courtyard level (V8.3); FF by the lift; thresholds 15 mm	met	The courtyard's step removed (V8.3); the family entrance: no step drawn (the ground −0.30 against the floor ±0.00 needs a ramp or a raised path: **not drawn**, to detail).
R-09	Safe bathrooms; no GF shower (owner)	Level showers in the suite, bedrooms 2 and 3; the powder room with a 1.5 m circle	partly	Grab-bar blocking not specified; the owner removed the ground-floor shower (3 Nov 2026): ageing in place relies on the lift; bedroom 1 without a shower on its floor (V-03).
R-10	Stair for life	`studies/structure.typ` stair page (flipped)	met	22 × 0.164 / 0.28, flights 1.21 clear, straight, both handrails assumed, headroom 2.22 (2.14 with the finish: V-13).
R-11	Family entrance from two shaded spaces, ≤ 10 m covered walk	`studies/site_plan.typ`: the covered parking is **outside the plot** on the farm; one uncovered bay inside	not met	The covered spaces are beyond the plot (walk ≈ 40 m on the driveway, uncovered); the bay inside is uncovered and 0.4 m from the majlis window (V-07); no canopy at the family entrance.
R-12	Men's majlis 20–24	GF men's majlis 46.77 m² clear	met	≈ 24 m of perimeter seating (V6 review); the serving route from the serving passage round the seats.
R-13	Large events elsewhere	—	met	The old majlis on the farm (decision X-01).
R-14	Guest facilities separate	Foyer, majaz, guest WC, trough, the serving passage	met	Own double door 1.40; the majaz partial (foyer 2.6 m: accepted, open item 13); one WC for 24 (V-15).
R-15	No sightline into family areas	The serving passage's two doors, the solid TV wall, the guest path on the driveway's west edge	met	Inside ✓; outside: the guest path never sees the garden (the 1.2 m wall), the north strip or the yard; the family's cars share the driveway (accepted).
R-16	Women guests via the family entrance	Women's majlis 28.24 m² off the lobby	met	Through the family lobby (the coat cupboard and the stair in view: accepted).
R-17	Closed heavy kitchen, hood with make-up air	Kitchen 23.99 clear, door to the dining, the laundry entrance open, the hood with an interlocked make-up inlet	partly	The laundry's entrance has no door (the kitchen is closed to living space ✓, open to the maid wing); the hood exhaust position conflicts with the outdoor counter (V-08).
R-18	Pantry ≥ 1.8 m	`east_wing.typ` tall storage	met	1.8 m run.
R-19	Outdoor kitchen by the service door, downwind	`site_plan.typ`: the counter y 8.3–10.1, the grill x 16.5–17.6	met	Distances checked (≥ 3 m from every opening, 13 m from the intake); **but** the hood exhaust beside the counter (V-08).
R-20	Yard screened from the outdoor-kitchen path	The screen at x 15.2, the gas at its north end, the bins north of the door	met	The workers' path never crosses the outdoor kitchen (checked); the bins near the maid's window (V-08).
R-21	Living + dining ≈ 53.5 m² clear	27.83 + 25.63 = 53.46 m²	met	A table for 10 and TV seating for 6–8 drawn in the presentation's furniture.
R-22	TV 85" on the solid west wall	The TV wall with the arches (V6.3)	met	Sofa 3.5 m from the screen; conduits to specify.
R-23	Courtyard: sitting, coffee, meals	28.35 m² level, 5 seats + the table, the TV, the water, the hose point	met	Play in the garden (R-30); no pergola.
R-24	Level floor, movable seating, the TV, the enclosure at the kitchen wall (3 Nov)	`studies/courtyard_layout.typ`, the 3D	met	9 study checks; the raised trough seen from the sink; the grille clear of the furniture; cushion storage (V-17).
R-25	Closed, filtered house	W1 / W2 / W3 (V7.0); the HRV 750 m³/h class with ePM1 ≥ 50 % in the malqaf base; the dust mode	partly	The concept is right; the HRV, the ducts and the pressurisation are the MEP engineer's (open item 9); airtightness and the window specification not written (V-04).
R-26	Owner's office ≥ 12 m²	13.96 m² clear, from the suite court	met	For the owner only; visitors in the majlis.
R-27	Gym ≈ 18 m²	19.92 m² clear, the treadmill zone, the double wall to bedroom 3	met	Rw ≈ 55 dB target (open item 8).
R-28	Master suite: tub + shower, quiet	Suite 55.6 m²; the bath over the kitchen	partly	Tub and walk-in shower ✓; the bath's drainage approach undecided (V-10); the kitchen below (noise through the depressed slab: wrap the pipes).
R-29	No pets	—	met	—
R-30	Garden round the house, play, family access	The south garden ≈ 289 m² round the palms; no garden door (owner)	partly	Garden on the south only (owner's decision); access from the kitchen's service door round the corner or from the guest side: no direct family access from the living (S-04 "not needed").
R-31	Solar PV provision	30 m² of roof zones, the conduits down the west shaft, the inverter and batteries in the utility room	partly	The utility room overfull (V-10); the structural load allowance to confirm.
R-32	Power cuts: backup (no generator)	The PV + batteries later; an essentials circuit	not verifiable yet	Until the batteries nothing runs in a cut: no water (V-05), no HRV, no lift.
R-33	Security: CCTV, alarm, intercom	`site_plan.typ` page 5: the gate intercom, 7 cameras, the alarm panel, keypads	met	Positions proposed; the electrical engineer designs.

R-34	Luxury in the living and the court-yard	The double-height living, the arcade and W3, the stone courtyard	**partly**	Light and sound not yet studied (V-19); the materials in the design language.
R-35	Recent decisions respected	Every decision of V6.3 … V8.3 present in the files	**met**	Checked against `data/decisions.csv`; the record studies are stale by design (V-18).
R-36	Fire safety at home level	`studies/fire_escape.typ`: alarms, escape casements, thumb-turns, the court a refuge	**met**	Met as the owner defined it; the residual risk of the open stair recorded (V-09).
R-37	Facade principle (no privacy from outside)	To be applied at the facade stage; the list of walls to open in `PROJECT_V8.md` §6	**not verifiable yet**	Nothing drawn yet; the parking bay against the north facade conflicts with it (V-07).

مصفوفة التتبع: من 37 متطلبا: 24 متحققة، 9 جزئيا، 1 غير متحقق (R-11: المواقف المظللة خارج القطعة وبلا مظلة للمدخل)، 3 لا يمكن التحقق منها بعد (الطاقة الاحتياطية، مبدأ الواجهات، المصعد). التحقق هنا يعني أن الملفات تبين المطلوب مرسوما أو مقررا، لا أن جهة رسمية اعتمدته.

2 · Verification · Saudi rules and global practice

For each check: the source cited; "not verified" where the rule's figure was not read from the code text or the municipal file, with what the designer of record must confirm. The files contain no calculation; a check line in a study proves the drawing's internal consistency only.

Rule / practice (source)	What the design shows	Result	Not verified / to confirm
SBC 1101 (residential, IRC-based) · habitable rooms ≥ 6.5 m², ceilings ≥ 2.13 m	Every bedroom ≥ 12.4 m² clear; ceilings 3.6 – 0.3 (ribbed) – 0.15 (gypsum) ≈ 3.1 m; under the drop beams B4 / B5 soffits +3.05 … +3.10 (gypsum below ≈ 2.9)	**met**	**Not verified** against the SBC text (the IRC R304 / R305 figures used).
SBC 1101 · light and ventilation (IRC R303: glazing ≥ 8 % of the floor, openable ≥ 4 %)	Bedroom 2: 1.3 × 1.2 ≈ 1.6 m² / 12.84 = 12 %; bedroom 3 ≈ 13 %; bedroom 1 ≈ 3.5 m² / 13.8 = 25 %; the kids' room 3.4 × 1.2 = 4.1 / 12.4 = 33 %; the women's majlis (north window 2.4 m + courtyard glass) ✓; the maid room 1.14 × 1.0 / 7.3 = 16 %; the family living: W2 only (2 × 1.3 m arched, floor to +6.6) ✓; the utility room, the powder room, the family WC: mechanical only	**met**	Mechanical ventilation (HRV) covers the internal rooms; the figures are from the window lists, the sill / head from `3d/params.yaml` (assumed).
SBC 801 / IRC R310 · escape openings in sleeping rooms (≥ 0.53 m² net, sill ≤ 1.12 m)	Every sleeping room has a hinged casement without a screen; the maid room's at sill +1.10; bedroom 1's on the cut wall 1.6 × 0.87	**met**	The net clear opening of each casement is to be confirmed with the window schedule (hinged, not top-hung).
SBC 801 / IRC R314 · smoke alarms interconnected, in every bedroom, outside each sleeping area, every storey; R315 CO	Drawn (S marks on both plans), mains with battery backup; a heat alarm in the kitchen; CO by the gas	**met**	—
SBC 801 · means of egress from a dwelling (one exit; the stair's width ≥ 0.9, rise ≤ 0.196, going ≥ 0.254, headroom ≥ 2.03)	One stair 1.21 clear, 0.164 / 0.28, headroom 2.14 with the finish; open to the riwaq	**met**	The open stair and the single route: a residual risk recorded (V-09). **Not verified**: whether the Civil Defense's villa checklist asks for an enclosed stair in a two-storey villa of this size.
SBC 701 · drainage: trap seals ≥ 50 mm, every trap vented, cleanouts at direction changes, water-sewer separation 1.5 m horizontal / 0.3 vertical, septic ≥ 15 m from a well (practice 30 m)	Principles stated and drawn in the plumbing study; the vents to the roof; the chambers at every bend; the separations checked on the site plan (assumed rules)	**met on paper**	The study is a proposal: **the plumbing engineer must size the pipes and confirm the falls (1 % main, 1.5 % branches)**; the septic's legal position **not verified** (V-06).
SBC 501 / ASHRAE 62.2 · dwelling ventilation ≈ 0.15 l/s·m² + 3.5 l/s per person; intake ≥ 3 m from exhausts and contaminant sources	The air study: ≈ 330 m³/h continuous, 705 with 24 guests, a 750 m³/h class HRV; the intake in the malqaf 13 m from the grill; the hood exhaust position (V-08)	**partly**	Figures assumed by the study; the MEP engineer designs (open item 9); the exhaust / intake / outdoor counter conflict is new.
SBC 601 / 602 · energy: envelope U-values, window-to-wall ratio, airtightness	Nothing in the files: no insulation, no glazing specification, no WWR calculation; the steel windows single-framed; no heating	**not met**	**Not verified**: the designer of record must run the SBC 602 path and add a heating concept (V-04).
SBC 301 / 303 / 304 · structure and foundations	Assumed sizes throughout; no soil report	**not verifiable**	V-02: the engineer of record and the geotechnical investigation.
MOMAH villa requirements (as reported July 2024): building ratio ≤ 75 %, two parking spaces for villas over 400 m², maid room ≥ 6.5 m² and ≥ 2.1 m, setbacks	Footprint 318.77 m² against the envelope 667.87 (≈ 48 %); the two covered spaces are outside the plot on the farm, one uncovered inside; the maid room 7.28 / 2.49 ✓; the setback margin 0.20 provisional	**partly**	**Not verified**: the Taif municipality's reading of a villa inside a walled farm (the parking's location, the plot boundary used for the ratio, the septic outside the wall); the setback (V-01).
Taif architecture guidelines (the Taif Municipality's heritage style guide: honey stone, mashrabiya, cornices)	The design language: cream plaster, honey stone at the plinth and corners, black steel divided lites, the arcade and the arched enclosure windows, a few mashrabiya	**met in spirit**	**Not verified**: the guide's actual clauses (colour palette, window proportions, roof parapets) were not read; the facade stage should check them with R-37 in mind.
ISO 21542 / ageing in place: doors 0.85 clear, corridors 1.2, turning 1.5, level showers, a lift	Doors 1.00 openings (two exceptions), the gallery 1.235–1.405, the powder room and the lift lobby with 1.5 m circles, level showers upstairs, the lift at construction	**partly**	V-11, V-12; no ground-floor shower by decision; grab-bar blocking not specified.
Space standards and furniture clearances (Neufert / Metric Handbook): beds with 0.6–0.75 m beside, wardrobes 0.6 deep, dining 0.9 round the table, WC 0.6 in front	The studies' furniture checks (bedrooms, the suite, the kitchen's two runs 2.94 m apart, the WC fronts 0.6, the courtyard's 0.8 in front of the grille)	**met**	The family link's two doors (V-12); bedroom 2's bed seen from its door (noted, accepted).
Daylight (BS EN 17037) and acoustics (practice)	No simulation; the clerestory sun study only; the gym's double wall specified	**not verifiable**	V-19.
Dust and indoor air (ISO 16890 ePM1 ≥ 50 %, positive pressure, a closed envelope)	Specified in the air study; the envelope closed by W1–W3; the courtyard open	**met in concept**	Airtightness and the window seals not written (V-04).
Plumbing maintainability and smells (BS EN 12056, practice)	The plumbing study's principles, the smells page, trap primers decided, the external stack behind cladding	**met in concept**	The suite bath's approach undecided (V-10).
LPG (NFPA 58 as used in Saudi practice): ≥ 1.5 m from openings, ≥ 3 m from ignition and drains	The cabinet 2.0 m from the service door and the kitchen window, 3.9 m from the grill, 4.8 from the manhole	**met**	To confirm with the Civil Defense (the site plan says so).
Security (practice: one controlled entry, cameras on every approach, an alarm with a panel in a service room)	One gate with intercom and camera, seven cameras, the panel in the utility room, keypads at both entrances	**met**	The electrical engineer designs.
Buildability (practice)	Walls on walls on the first floor except y 3.0, y 9.2 (new), y 11.0, the suite walls, the gym nib and the office wall (noted in the structure study); the 1.55 m projection on three columns	**partly**	V-02; the ribbed slabs with a depressed zone and hidden / drop beams need the engineer's confirmation before the ceiling heights are promised.

القواعد والممارسة: المتطلبات السعودية للمساحات والإضاءة والهروب وأجهزة الإنذار متحققة على الرسومات؛ السباكة والتهوية مبادئ صحيحة تنتظر تصميم المهندس؛ كود الطاقة (SBC 602) والتدفئة غائبان؛ الإنشاء والتربة غير محققين؛ اشتراطات البلدية (المواقف داخل القطعة، الارتداد، خزان التحليل خارج السور) تحتاج تأكيدا من البلدية. كل بند لم يُقرأ من مصدره مكتوب «غير محقق».

3 · Verification · consistency across the files

`tools/check.py` compiles everything, compares the two studies' rooms with the CSVs, the DXFs' areas with the CSVs and rebuilds the 3D with its own checks: every geometric link is machine-checked. The mismatches below are therefore in the **texts and records**, not in the geometry.

- **Room names and sizes:** `data/rooms_gf.csv` / `rooms_ff.csv` ↔ `guest_wing.typ`, `east_wing.typ`, `ff_family.typ`, `ff_suite.typ` ↔ the DXFs ↔ the 3D: **no mismatch** (check.py sections 3, 4, 6). The presentation's schedules and `PROJECT_V8.md` §3 are generated from the same data.
- **Doors and windows:** the ground-floor doors come from `presentation.typ` (gf-doors), the first-floor ones from the two studies, the heights from `3d/params.yaml`; the 3D flags any height without a door: **none**. One text mismatch: `docs/AIR_SYSTEM.md` described the grille at the old height until V8.3 (corrected in this release).
- **Stair and lift:** `studies/structure.typ st` → the 3D (direction-aware) → the fire study: consistent; `studies/stair_flip.typ` still shows the **old** family bath and the pre-flip routes as its record (V-18).
- **Plumbing clusters and stacks:** `studies/plumbing.typ` rounds 1–2 ↔ `ff_family.typ` drains ↔ `guest_wing.typ` ↔ the site plan's chambers: consistent after V8.3; the round-1 scenario B still names the GF shower P1 as data (withdrawn in the text).
- **Roof ↔ plumbing:** the roof reserves a **water tank zone** (R10–R3); the plumbing study's water scheme has **no roof tank** (V-05): a concept mismatch to decide.
- **Site plan ↔ plans ↔ 3D:** the counter, the grill, the screen, the gas, the bins and the paths are the same numbers in `site_plan.typ`, `design_gf.typ` and `3d/character.yaml` (checked by hand at V8.3); the site plan's parking bay vs the north windows (V-07) and the hood vs the counter (V-08) are **design** conflicts the checks did not cover.
- **Open items:** `data/open_items.csv` ↔ `OPEN_ITEMS.md` ↔ the presentation sheet 10: numbered in the same order (checked); the review status rows updated for V8.3.
- **Decisions log:** 57 rows; the V8.1-labelled rows mix the tower removal (V8.1) with items applied in V8.2 (relabelled at V8.2) — a labelling, not a content, mismatch (V-18).
- **Record studies that no longer describe the house:** `stair_flip.typ` (the family bath), `south_wing.typ` (the stair tower and the stair room in its sections), `gf_shower.typ` (closed), `bedroom1_bath.typ` pages 1–2 (option A as adopted, now reopened), `air_system.typ` ("the benches" in one table head): each carries a header note; none has a page-1 banner (V-18).

الاتساق: الروابط الهندسية بين الملفات تُفحص آليا ولا تعارض فيها؛ التعارضات الموجودة في النصوص والسجلات القديمة (دراسة قلب الدرج، دراسة الجناح الجنوبي، دش الأرضي، خزان السطح بين دراسة السطح ودراسة السباكة) وفي قرارات التصميم الأخيرة (الموقف أمام النوافذ الشمالية، مخرج الشفاط بجوار المطبخ الخارجي).

4 · Validation · does the house work for this family?

Each scenario walked on the V8.3 plans (`design_gf.pdf`, `design_ff.pdf`, the studies). ✓ works · awkward · ✗ fails.

Scenario	Walk-through on the plans: what works, what is awkward, what fails
A normal school day (5–6 people, the maid)	**06:00** the maid from her room through the laundry to the kitchen (no family crossed) ✓; **06:30–07:30** the children wash: bedroom 3's en-suite ✓, bedroom 2's en-suite ✓, **bedroom 1 and the kids' room share the family WC — no shower on their floor** ✗ (V-03); the parents in the suite ✓; breakfast in the dining (the kitchen door) ✓; **07:45** out through the lobby to the car in the north bay (0.4 m from the majlis window , V-07) or the farm parking (40 m, uncovered); **14:00** lunch, the courtyard in winter ✓; homework in the kids' room (3 desks) ✓; **evening** the family living on the courtyard, the TV wall in the living ✓; **night** the parents' route bedroom → suite bath 0 m ✓; bedroom 1 → the family WC 15 m through the family living (night lights) .
Weekly men's guests while the family is home	The guests arrive by the farm parking and the stone path to the double door; the foyer's majaz (partial, accepted) → the majlis; the family's rooms, the courtyard and the cars are never seen ✓; the trough and one WC for 20–24 (V-15); serving: the hospitality kitchen → the riwaq → door 1 → the serving passage → door 2 → the host collects (the two doors never open together: a rule, not a device); the food route from the main kitchen through the dining and the riwaq (accepted X-01) ✓; the family upstairs or in the family living hears nothing (the majlis under the gym / court / office: the gym's double wall is to bedroom 3, the majlis's ceiling to the gym is a ribbed slab an evening treadmill over a majlis in session: specify the floating floor).
Close relatives visiting the family	In by the family entrance → the lobby → the riwaq → the living (the arch) or the courtyard (the riwaq's sliding door) ✓; the women's majlis for a larger group ✓; the **powder room** from the riwaq (two basins, a wall-hung WC, 7.6 m²) ✓ luxury; children to the garden through the kitchen's service door round the corner (no door from the living, by decision); the courtyard's 5 seats + the living's 6–8 ✓.
The maid's day	Shopping in: gate → the boundary-wall path → the lane-side strip → the yard → the service door → the kitchen (never the family's rooms) ✓; rubbish out: the kitchen → the service door → the bins 2 m north ✓ (the bins 1 m from her own window , V-08); laundry: the machines beside her room, the sink; **no drying area** ✗ (V-16); cooking: the heavy kitchen with the pantry; her bath 3.0 m² with a 0.70 door ; her window an escape casement at sill 1.10 ✓.
A dust storm	Close: W1 / W2 / W3 and the riwaq's and living's glass are already the envelope ✓; the suite court's three doors ✓; the malqaf damper closes (spring-return) and the breeze channel is sealed by its covers ✓; the HRV on dust mode (ePM1 ≥ 50 %, positive pressure) ✓; the courtyard collects dust on its level floor (a hose point, the linear drains with silt baskets) ✓; the cushions to the hospitality kitchen (V-17); the outdoor kitchen and the grill uncovered (a canopy optional) .
A power cut in rain	The lights: no backup until the batteries ✗; the alarms on their batteries ✓; the gas valve fails closed ✓; the lift: battery lowering to specify ; the HRV stops (the house closed, the courtyard breeze by the damper only) ; **water: the booster pump stops and there is no roof tank** ✗ (V-05); the rain itself: the courtyard's linear drains and the overflow work by gravity ✓, the roof outlets ✓, the septic by gravity if the inlet is low enough (else the pump station stops ✗, V-06).
A fire at night starting in the kitchen	The heat alarm in the kitchen and the interconnected smoke alarms wake everyone ✓; the kitchen's door to the dining is a plain door (no closer) , the living / dining is one room, the living's arch has no door: smoke reaches the riwaq and the open stair quickly ✗ (V-09); the parents: the suite bedroom's east escape window (y 15.0–18.6) ✓ or the family link → the gallery → the stair ; bedrooms 2 and 3: the stair (if clear) or their west escape windows onto the driveway where the ladder stand is (fire study P2) ✓; bedroom 1 and the kids' room: the cut-wall and north escape windows ✓; the maid: her casement at +1.10 ✓; the gas detector and the valve ✓; extinguishers at the stair and in the kitchen ✓.
An elderly parent staying a week	The lift from the lobby to the gallery ✓; bedroom 2 (a level en-suite shower, 12.8 m²) is the right room (a child moves out) ✓; bedroom 1 would be wrong (no shower, V-03) ✗; the family WC from the gallery ✓; the gallery 1.235 at the columns ; the courtyard level ✓, the garden by the kitchen's service door only ; no ground-floor shower by decision (a week with the lift ✓; a broken lift ✗: an accepted risk, R-09).
A roof maintenance visit	The technician enters through the house: the lobby → the stair / lift → the gallery → the gym → the suite court (the gym's and the court's doors; the suite's bedroom door closed) → the ladder behind its panel → 3.7 m up with a fall-arrest rail → the gate in the court guard → the walkways to the HRV doors, the condensers, the tank zone and the PV; tools by hand up a vertical ladder (V-14); the attic hides nothing from him; the ladder's foot in the suite court means the suite is entered by workers (accepted at V8.1).
A hot summer day / a cold winter night in Taif	**Summer:** the courtyard's west half under the projection's soffit (shade), the open strip in the sun until mid-afternoon (no pergola by decision), the breeze path on demand, the AC; the south wing's suite court in full sun (the mashrabiya screen shades the south side) ; the west bedroom windows in deep reveals ✓. **Winter night (≈ 0–5 °C):** the courtyard glass W1 / W2 / W3 single-framed steel (no thermal break specified), the open-sky courtyard against the living's 6 m glass wall, **no heating system in the concept** ✗ (V-04); the clerestories bring the winter noon sun deep into the living ✓.

A day in the life · route maps

Who	Routes (CSV frame rooms) and the distances that matter
Parents	Suite bedroom ↔ suite bath 0 m ↔ dressing; the suite door → the family link → the family living (W2) → the gallery → the stair (or the lift) → the lobby → the kitchen / dining (≈ 25 m) → the courtyard; the office from the suite court (the owner only); the car by the lobby. Awkward: the family link's two doors.
Older child (bedroom 3)	Bedroom 3 ↔ its en-suite 2.7 m; the gallery → the stair → the lobby → the dining; the gym 10 m along the gallery; the suite court for air. Works.
Young child (bedroom 2 / the kids' room)	Bedroom 2 ↔ its en-suite 2.2 m ✓; the kids' room (desks, gaming) → the family WC 8 m, **no shower on the floor** (V-03); the family living on the courtyard; the garden by the kitchen's service door round the corner (no living door) .
Bedroom 1's user (a child or a guest)	Bedroom 1 → the family link → the family living → the gallery → the family WC 15 m (night lights); a shower only in bedroom 2's, bedroom 3's or the suite's bath ✗ (V-03); the escape window on the cut wall.
Maid	Her room ↔ her bath 0.70 door; the laundry → the kitchen → the dining (serving) or the service door → the yard (bins 2 m, the gas, the outdoor kitchen); shopping from the gate along the boundary wall ≈ 35 m; never through a family room ✓.
Guest (men's majlis)	The farm parking → the stone path on the driveway 15.5 m → the double door → the foyer (shoes, the majaz) → the majlis; the trough and the WC off the guest passage; served through the two-door passage; never sees a family space ✓. Women guests: the family entrance → the lobby → the women's majlis ✓ (through the family lobby).

التحقق من الصلاحية: يوم الدراسة والضيوف الأسبوعيون والأقارب ويوم الخادمة والعاصفة الترابية وزيارة الصيانة تعمل على المخططات، مع ملاحظات. يفشل البيت في: دش غرفة النوم 1 وغرفة الأطفال في دورهما؛ الماء والإضاءة عند انقطاع الكهرباء قبل تركيب البطاريات؛ انتشار الدخان من المطبخ إلى الدرج المفتوح؛ التدفئة في ليالي الطائف الباردة.

5 · Open decisions and pending items

Decisions for the owner

- **Bedroom 1's shower** (open item 16): B4-b with its two failed criteria (a 7.3 m² L bedroom, the doors at a corner), a shower back in the family WC at bedroom 2's expense, bedroom 1 as the guest room sharing bedroom 3's en-suite through a second door, or accept the walk; the kids' room as a future bedroom has the same question.
- **The suite bathroom's drainage approach** (a: every drain in the depressed slab / b: the east service wall and the tub's plinth; the review recommends b) (open item 10).
- **The water system:** a roof tank (the zone exists) for gravity water in a cut, the booster, the heaters (two cylinders or a roof heat pump), any greywater line (the owner's question; none is drawn), the utility room's door moved south (open item 10; V-05, V-10).
- **The batteries' room:** the utility room (overfull) or the store under the half landing with a fire-rated door (V-10).
- **The north parking bay** against the north facade, or the farm parking only (V-07); a canopy at the family entrance (R-11).
- **The outdoor kitchen vs the hood exhaust** (V-08): the counter north of the service door or the hood outlet raised.
- **Heating** and the window specification (V-04): the scope of the MEP brief.
- **The clerestories' smoke-vent actuators and a door at the living's arch** (V-09).
- **The HRV and malqaf details** that remain the owner's: the assist fan's switch, the damper's control, the dust mode's trigger (the MEP engineer proposes).

Engineer items

- **Structural (items 2–7):** the soil investigation; X5 (the south wing on the men's majlis roof); the edge beam and the three columns; the stair trimmers (0.30 option); X21 (the first-floor service wall, B14, the powder room's service wall); the depressed suite-bath zone; the roof tank's load; the PV load; the lift pit.
- **MEP (items 9–11):** the HRV, the risers, the malqaf's damper and fan, the kitchen make-up air (and its position: V-08); the plumbing design (the falls, the pipe sizes, the stacks, the pump station if the septic's inlet is high, the heaters, the roof tank, the trap primers); heating and the SBC 602 envelope; the electrical (the panel, the batteries' room, the essentials circuit, the cameras, the intercom, the night lights).
- **The lift supplier (item 14):** two quotations, the larger shaft, the headroom ≤ 3.40 m, the battery lowering.
- **The Civil Defense:** the villa checklist (access, water, alarms), the gas cabinet, the escape windows' net openings.

Survey items

- The legal survey of the plot and the setback lines (item 1, V-01); the street level at the gate; the pegs M1–M4, corner C1, peg R.
- The septic (position outside the wall, inlet level, capacity, condition), the existing tank east of peg R, the well, the old house's floor (item 18); the municipality's consent (item 19).

Detail-stage items

- The facades under R-37 (the list of walls to open; the Taif guidelines; shading and glazing with SBC 602) (item 17).
- The living's acoustics (E-07), the daylight simulation (item 15).
- The bathroom interiors (the suite bath's fittings and the wall-drain channel system; the powder room's stone vanity; bedroom 2's and 3's en-suites; the family WC's screen wall), the kitchen's joinery, the courtyard's furniture and the enclosure's stone.
- The thresholds (15 mm) and the family entrance's ramp from −0.30 to ±0.00; the drying area; the cushion store; the gallery's columns' sleeves.

القرارات المفتوحة: دش غرفة النوم 1؛ أسلوب صرف حمام الجناح؛ نظام الماء (خزان السطح، السخانات، الخط الرمادي)؛ غرفة البطاريات؛ موقف السيارة أمام الواجهة الشمالية؛ المطبخ الخارجي ومخرج الشفاط؛ التدفئة والنوافذ. وللمهندسين: التربة والإنشاء (X5, X21)، التهوية والسباكة والكهرباء، مورد المصعد؛ وللمساح: الحدود والارتداد وخزان التحليل والمناسيب.

6 · Comparison with the V6 review (30 Oct 2026)

- **Resolved since V6:** the three criticals are two-thirds closed: Y-01 (the courtyard envelope: W1 / W2 / W3, V7.0) and A-01 (the lift at construction, V6.3); S-03 (the setback) is **still open** (V-01). Of the 27 majors, 20 are resolved on the files (the TV wall, the maid room, the gallery width, the door widths, the courtyard layout and its safety, the air system, the fire strategy, the roof access, the office, the site plan's parking, guest approach, garden, yard, PV, generator, security, the food route, the ground-floor bathrooms by decision), 6 partly (G-02 daylight, G-03 MEP, T-01 structure, E-02 / E-03 MEP and plumbing, S-08 the septic and tanks), 2 open (T-03 soil, E-07 acoustics).
- **What improved most:** the house became closable and filterable (R-25), the roofline and the roof access were simplified, the site is planned, the plumbing has a strategy, every bedroom but one has its own bathroom, the courtyard is level and furnished.
- **What is new since V6 (not in that review):** the heating / energy gap (V-04: the V6 review flagged the open envelope, not the missing heating); the power-cut water (V-05, created by removing the generator); bedroom 1's shower (V-03, created by the 3 Nov split); the north parking bay against the windows (V-07) and the hood beside the outdoor counter (V-08), both created by the site decisions; the septic outside the wall (V-06, the owner's information); the record-keeping drift (V-18).
- **Severity trend:** V6 had 3 critical / 27 major / 25 minor-suggestion findings; this interim review has 2 critical / 8 major / 8 minor / 2 suggestions. The two criticals are the same two unresolved external dependencies (the survey, the engineer); the majors are fewer and more specific.

مقارنة بمراجعة V6: حُلت اثنتان من ثلاث مسائل حرجة (غلاف الفناء والمصعد) وبقي الارتداد؛ حُل معظم الكبيرة (20 من 27)؛ الجديد: التدفئة والطاقة، الماء عند انقطاع الكهرباء، دش غرفة النوم 1، موقف السيارة ومخرج الشفاط، خزان التحليل خارج السور.

7 · Findings register

Every finding: ID · severity · location · what is wrong or could be better · evidence · the requirement / rule / practice · the recommended fix and its effect on other spaces. Sorted by severity, then by the top-10 order.

V-01 · critical · **Site · the setback margin**
What: The nearest outer wall face stands 0.20 m from the lane-side setback line (0.80 m from the north street line), a **provisional** margin set in V6.3 without the official survey. A survey difference of a few centimetres puts the wall in the setback; the whole plan is set out from peg M1 and the envelope of `base.typ`, neither confirmed. Three versions later this is still open (V6 review S-03, critical).
Evidence: `design_gf.typ` env-gap 0.20 (computed against `base.typ` env); `OPEN_ITEMS.md` item 1; `studies/site_plan.typ` "every position proposed, to confirm with the official survey".
Rule / practice: MOMAH villa requirements: no building in the statutory setback; the exact Taif setback dimensions behind the envelope are **not verified** against the municipal file.
Fix and effect: Commission the legal survey before any further geometry work; if the margin is wrong, move the house south (the garden has ≈ 7 m) or trim the maid wing. Effect: everything downstream (the DXFs, the 3D, the site plan) is parametric on the shift, so a move is cheap now and very costly after the engineers start.

V-02 · critical · **Structure · unverified concept, no soil report**
What: Every structural size is assumed: the column grid, the 1.55 m west projection on an edge beam and three columns, the ribbed slabs, the men's majlis roof carrying the gym, the suite court terrace and the office (X5, "revised, improved, **not verified**"), the stair trimmers, the new B14 condition (X21). No soil report exists; the pads, the lift pit, the breeze channel in the slab and the soakaways are drawn on an assumed bearing stratum. The design has progressed to interiors and plumbing on a structure nobody has checked.
Evidence: `studies/structure.typ` X5, X13–X15, X21; `docs/STRUCTURE_STUDY.md`; `OPEN_ITEMS.md` items 2–7; the 3D's "assumed" flags.
Rule / practice: SBC 301 / 304 (loads, concrete): the engineer of record must size every element; a geotechnical investigation precedes the foundation design (SBC 303). **Not verified**: nothing in the files is a calculation.
Fix and effect: Appoint the structural engineer now with the soil investigation; freeze nothing that depends on beam depths (the ceiling heights under B4 / B5 / B6, the 2.14 m stair headroom, the 0.15 depressed suite bath zone) until the first sizing round. Effect: possible deeper beams under the first-floor west rooms and the suite walls; the stair trimmer option 0.30.

V-03 · major · **FF · bedroom 1 (and the kids' room) without a shower**
What: V8.3 turned the family bath into a WC and gave bedroom 2 the shower. Bedroom 1 (and the kids' room when it becomes a bedroom) now has **no shower on its floor**: its users must enter bedroom 2's or bedroom 3's en-suite or the suite. The hotel-layout en-suite B4 cannot fit the wedge room (its study page 3); the ground floor has no shower by decision. The house therefore has a bedroom whose daily route to a shower crosses another child's bedroom: a validation failure for a 5–6 person family with children of mixed ages (R-01, R-09).
Evidence: `data/rooms_ff.csv` (Family WC (gallery) 2.75 × 1.3; Bedroom 2 bath (en-suite)); `studies/bedroom1_bath.typ` page 3 (B4 fails criteria 2 and 3); `studies/ff_family.typ` "no shower for them on this floor"; decision "Bedroom 1: en-suite … studied, NOT applied".
Rule / practice: R-01, R-09 (a shower reachable for every resident without crossing another's room: practice, Neufert / Metric Handbook household planning); SBC 1101 does not require it, so this is a validation, not a code, failure.
Fix and effect: Decide now, before V9.0: (a) B4-b for bedroom 1 with its 7.3 m² L bedroom (the owner accepts the two failed criteria), (b) a shower back in the family WC by taking 0.6 m from bedroom 2's bath (bedroom 2's shower 1.0 × 1.0: tight), (c) bedroom 1 declared the guest / older-child room that shares bedroom 3's en-suite through a second door (bedroom 3's bath door is at x 2.2 inside bedroom 3's entry, far from bedroom 1), or (d) accept the walk. Effect: (a) bedroom 1's furniture and the laundry's corner drop; (b) bedroom 2's bath; (c) none.

V-04 · major · **Services · no heating, no energy-code envelope**
What: The services concept covers ventilation (HRV), cooling (AC condensers on the roof) and the courtyard breeze; **heating is absent**. Taif at ≈ 1,700 m has winter nights near 0–5 °C; a courtyard house with large north-facing courtyard glass (W1 / W2 / W3, single steel-framed divided lites with no thermal specification) and an open-sky courtyard needs a heating strategy (reversible heat pumps, underfloor heating in the living and the baths) and an envelope specification (insulation, glazing U-value, thermal bridges at the steel windows, airtightness). None exists.
Evidence: `docs/AIR_SYSTEM.md` (HRV, AC condensers, no heating); `studies/courtyard_enclosure.typ` (W1–W3 as steel divided-lite windows, no U-value); no energy section in any file; `docs/REQUIREMENTS.md` R-25 (dust) only.
Rule / practice: SBC 601 / 602 (energy conservation: envelope U-values, window-to-wall ratio limits, airtightness) — **not verified**: the designer of record must run the SBC 602 prescriptive or performance path; ASHRAE 55 comfort; practice: Taif's winter design temperature.
Fix and effect: Add a heating concept to the MEP brief (reversible VRF or split heat pumps; underfloor heating in the living, the courtyard-facing rooms and the bathrooms), specify thermally broken steel windows and the wall / roof insulation before the facade stage; check the WWR of the courtyard glass against SBC 602. Effect: the roof plant zone (heat pumps), the slab build-up (underfloor heating thickness: the finish 0.08 assumed), the window frames' depth.

V-05 · major · **Services · water and power in a power cut**
What: The owner removed the generator (V8.2); the PV batteries are a later installation. Until then a power cut stops the booster pump (the only water supply is the ground tank 38 m away through a pump), the HRV, the lift and the assist fan. The roof study reserves a **water tank zone** between R10 and R3, but the plumbing study's water scheme (tank → booster pump → manifolds) has **no roof tank**, so there is no gravity water in a cut and the "water from the roof tank" scenario cannot be walked. The lift's battery lowering, the alarm panel's battery and the gas valve's fail-safe are stated; the water is not.
Evidence: `studies/roof.typ` (tank zone reserved); `studies/plumbing.typ` principles (booster pump, no roof tank); `studies/site_plan.typ` page 4 "no backup during power cuts"; decision "no generator (E-05)".
Rule / practice: SBC 701 (water supply continuity is not mandated, but a dwelling without water in a cut is a validation failure); R-32 as changed; practice: a roof break tank with a float valve gives 1–2 days of gravity water.
Fix and effect: Decide the water system now: a roof tank (≈ 2 m³, the zone exists) fed by the booster, gravity to the fixtures with a small pressure set only for the showers; an essentials circuit sized for the batteries (pumps, fridges, the HRV, lights, the lift lowering). Effect: the roof structure (R10 / R3 take ≈ 25 kN), the attic's hiding height (1.40 m: a 2 m³ tank is ≈ 1.2 m high: check), the plumbing risers in the west shaft.

V-06 · major · **Site · the septic and the long foul main**
What: The existing septic lies **outside the farm wall near corner C1**, under the street side: its legal status for a new house, its inlet level, its capacity and its condition are unknown. The east main runs 36 m north and west with its inlet computed at −1.19 below the house floor on assumed falls; if the surveyed inlet is higher, a sewage pump station at the gate is needed, with all its maintenance and power dependence (V-05). The foul drains also cross under the boundary wall and run in the driveway under the guest path.
Evidence: `studies/site_plan.typ` page 4 (inverts, "a pump station if higher", "the municipality's consent"); `OPEN_ITEMS.md` items 18, 19.
Rule / practice: SBC 701 (drain falls 1–2 %, chamber spacing, the separation from the water pipe 1.5 m); Saudi municipal practice (a septic serving a new villa: approval, the 30 m well distance); **not verified**: the municipality's rule on a septic outside the plot.
Fix and effect: Survey the septic first (position, inlet level, size); ask the municipality; if a pump is needed, consider a new septic / treatment unit inside the farm south of the house where the levels work by gravity. Effect: the whole foul route, the gate chamber, the driveway works.

V-07 · major · **Site · the north parking bay against the north facade**
What: The kept family car bay stands 0.4 m from the house's north wall, directly in front of the women's majlis north window (x 5.4–7.8) and under the kids' room's north window. A parked car 0.4 m from a window gives the room a bumper view, exhaust and heat, and it contradicts the new facade principle (R-37) that wants the north facade **opened** for soft north light. The site plan also makes the north strip "a calm family space" and parks a car in it.
Evidence: `studies/site_plan.typ` park-opt (x 5.2–10.2, y 0.9–3.4, 0.4 m off the wall); the women's majlis window `presentation.typ` gf-windows ((5.4, 0), (7.8, 0)); R-37 and the "walls to open" list; decision "the north-strip parking kept".
Rule / practice: R-37; practice: ≥ 1.5–2 m between a parked car and a habitable window, or a planted buffer; MOMAH parking rules (2 spaces for a villa over 400 m²: the covered spaces are **outside the plot**, on the farm — **not verified** whether the municipality counts them).
Fix and effect: Move the bay east along the strip (x 10–15, in front of the maid wing's cut wall, where the windows are small and obscured) or drop it and keep the farm parking; keep a 1.5 m planted strip at the majlis window. Effect: the swept path (the strip is 4–4.6 m deep: a bay further east needs the same 3-point manoeuvre), the workers' route along the boundary wall.

V-08 · major · **East yard · the hood exhaust beside the outdoor counter; the bins at the maid's window**
What: The outdoor counter and sink now sit under the kitchen window at y 8.3–10.1; the hood's exhaust outlet and the make-up air inlet are on the same wall at y 10.2–11.2, 0.1 m from the counter's south end. Cooking fumes and grease exhaust blow across the outdoor kitchen, and the make-up inlet draws the outdoor kitchen's smoke into the house. The bins (y 5.3–6.1) stand ≈ 1 m from the maid room's escape window on the cut wall (y 3.9–4.9 CSV).
Evidence: `studies/site_plan.typ` counter (13.45–14.05, 8.3–10.1), hood-out (13.311, 10.2–11.2), bins (13.7–14.5, 5.3–6.1); `studies/east_wing.typ` (the hob and hood on the east wall y 10.2–11.2); the site plan checks measured the gas against the hood but not the counter.
Rule / practice: SBC 501 / IMC 501.3 (exhaust outlets ≥ 3 m from property lines and ≥ 0.9 m from openings; practice: not over a work surface); ASHRAE 62.2 (intake separation from contaminant sources ≥ 3 m); R-19 / R-20.
Fix and effect: Either move the counter north of the service door (y 5.8–7.0, swapping with the bins: the bins then go south, screened) or raise the hood outlet to the band (+3.4) with a vertical discharge and move the make-up inlet to the cut wall's north part; move the bins ≥ 3 m from the maid's window. Effect: the yard plan, the hood duct (a 2 m rise inside the kitchen's east wall zone), the gas pipe route.

V-09 · major · **Fire · one open stair in the riwaq, smoke from the kitchen**
What: The only stair is open to the riwaq (no enclosure, no fire door, by the owner's home-level decision). A night fire in the kitchen reaches the dining and the double-height living (one room, no door between the living and the dining), the riwaq through the living's arched entrance (no door) and the stair well in minutes; the first floor's alternative exit is the escape window of each sleeping room (a 1.6 m drop from the sill to the ground is the drop from a first-floor window: ≈ 4.5 m to the ground). The clerestories vent the living's smoke only if someone opens them. This is the owner's accepted residual risk; the review records that the strategy relies entirely on early detection and on the windows.
Evidence: `studies/fire_escape.typ` (home-level strategy, one stair + escape windows, no fire doors); `design_gf.typ` (the living's arched entrance 1.40, no door; the kitchen's door to the dining); the 3D (the stair open on x 2.75, y 2.3–6.55).
Rule / practice: SBC 801 / IRC R314 (interconnected smoke alarms: met), R310 (escape openings: met), R311 (one exit from a dwelling: permitted); practice (BS 9991 / NFPA 101 for a two-storey house with an open stair: a protected route or a second stair is recommended, not required).
Fix and effect: Keep the decision but harden it: a heat alarm in the kitchen (drawn), a **closer on the kitchen's dining door** and a door at the living's arch is the cheapest protection of the riwaq and the stair; an escape ladder at bedroom 1's and the kids' room's windows; the clerestories on an automatic smoke-vent actuator linked to the alarms. Effect: the arch gets a glazed door (the design language accepts black steel); the clerestory actuators (open item 14 closed as openable: add the actuator).

V-10 · major · **Suite bath over the kitchen · the drainage approach undecided; the utility room overfull**
What: The one wet room over a habitable room is the suite bath over the kitchen. Two approaches are drawn ((a) every drain in the 0.15 depressed slab, the WC and the tub at the limit of the 0.14 m pipe zone; (b) an east service wall with the tub on a plinth, one local opening left); the owner has not decided, and the kitchen's ceiling, the hood zone and the east stack's branches depend on it. The utility room (2.5 × 2.5 axes) must hold the shaft, the main panel, the inverter and battery cabinet, the heater, the manifold cabinet, the filter, the booster connection, the alarm panel and the recorder: ≈ 4.3 m² of free floor, flagged small by the plumbing study itself.
Evidence: `studies/plumbing.typ` pages 7–9 (a / b), page 5 (the utility room, "small: reported"); decision "the suite bathroom's approach: NOT decided".
Rule / practice: SBC 701 (traps, vents, access to cleanouts: both approaches comply on paper); SBC 401 (an electrical room with batteries: clearances 0.9 m, ventilation, fire separation: **not verified**); practice: a lithium battery cabinet not in a wet room.
Fix and effect: Decide (b) before V9.0 (the review agrees with the study: accessible, nothing in the kitchen ceiling); move the batteries to the store under the stair's half landing with a fire-rated door, or enlarge the utility room by 0.5 m into the hospitality kitchen. Effect: the hospitality kitchen's counter (2.5 → 2.0 m), the stair store.

V-11 · minor · **GF · door widths below the accessibility target**
What: The guest WC's door is a 0.90 opening (≈ 0.75 m clear) and the maid bath's 0.70: below the 0.80 m minimum of R-07 and ISO 21542. Both were recorded as exceptions in V6.3 and stand.
Evidence: `studies/guest_wing.typ` doors; `docs/REQUIREMENTS.md` R-07.
Rule / practice: ISO 21542 (0.85 clear), R-07 (0.80 min).
Fix and effect: Accept (the guest WC and the maid bath are not on the ageing-in-place route) or widen the guest WC to 1.0 by taking 0.1 from the guest basins. Effect: none elsewhere.

V-12 · minor · **FF · the family link and the gallery at the columns**
What: The family link is 1.2 × 1.2 on axes (≈ 1.0 m clear) and takes bedroom 1's door (opening into bedroom 1) and the suite door (opening into the suite lobby): two people cannot pass. The gallery is 1.405 m clear but 1.235 at the column sleeves. Both meet the 1.0 m minimum of R-07, not its 1.2 m target.
Evidence: `studies/ff_family.typ` routes table; `data/rooms_ff.csv` Family link.
Rule / practice: R-07; ISO 21542 (1.2 m corridors, 1.5 m passing places).
Fix and effect: Accept; or widen the link to 1.5 by moving bedroom 1's door 0.3 m north on x 10.15 (the bedroom loses nothing). Effect: bedroom 1's wardrobe wall.

V-13 · minor · **Stair · headroom with the finish**
What: The flipped stair's headroom is 2.22 m to the structural trimmer at tread 6 and **2.14 m with the finish** in the 3D; the owner's R-10 asks ≥ 2.10. It passes by 4 cm on assumed trimmer depths (0.20 × 0.40); the engineer's 0.30 trimmer option gives 2.32.
Evidence: `studies/structure.typ` X15; `3d/check_3d.py` headroom check (2.14).
Rule / practice: SBC 1101 / IRC R311.7.2 (2.03 m): met; R-10 (2.10): met narrowly.
Fix and effect: Ask the engineer for the 0.30 trimmers at the first sizing. Effect: the first-floor opening's edge beams 0.10 deeper (hidden in the floor zone).

V-14 · minor · **Roof · maintenance by a vertical ladder**
What: Every roof visit (AC service, PV cleaning of 30 m² of panels, the HRV filters, the tank) climbs a 3.7 m vertical ladder in the suite court with tools and parts, through the suite's court. The owner chose it over the stair tower for cost and appearance (V8.1); the review records the residual: the frequency of roof visits (monthly PV cleaning in a dusty climate) makes the ladder the house's most-used piece of fall-risk equipment.
Evidence: `studies/roof.typ` page 2; decision "roof by a maintenance ladder".
Rule / practice: SBC 801 / practice (a fixed ladder with a fall-arrest rail: permitted for maintenance access); the specialist's confirmation is open (item 4).
Fix and effect: Keep; specify the fall-arrest system and a hoist point at the ladder's head for tools; put the PV cleaning tap and the hose on the roof. Effect: none.

V-15 · minor · **Guests · one WC for 24 men; the guest path beside the cars**
What: The guest side has one WC and a three-tap trough for 20–24 weekly guests (R-13 says no second WC is required; practice would give two for that number). The guest path is a 1.2 m band on the driveway, shared with the family's cars for 15.5 m.
Evidence: `studies/guest_wing.typ`; `studies/site_plan.typ` path-guest.
Rule / practice: R-12 / R-13 / R-14; practice (1 WC per 12–15 male guests in a majlis).
Fix and effect: Accept (the owner's R-13) or add a second WC in the guest passage's north end (1.25 × 1.4); a kerb or bollards between the path and the drive. Effect: the guest passage 3.9 → 2.5 m long.

V-16 · minor · **Maid's day · no drying area, the maid bath door**
What: The laundry has the machines and a sink but no drying line; the roof is reached by the ladder, the yard is the bins' and the gas's; the maid bath's door is 0.70 m. The maid's shopping route (gate → boundary path → yard → service door → kitchen → laundry → maid room) works and never crosses the family.
Evidence: `studies/east_wing.typ`; `studies/site_plan.typ`.
Rule / practice: R-02; practice (a screened drying area ≥ 3 m² near the laundry).
Fix and effect: A retractable drying line in the service yard behind the screen (y 11.5–13, the screened part) or a dryer only (drawn). Effect: the yard plan.

V-17 · minor · **Courtyard · the furniture in a storm, the sofa on the walkway's edge**
What: The movable seating is outdoor-rated but its cushions must be stored before a dust storm or rain; the study names the hospitality kitchen's cupboard (2.5 × 2.5 room already holding a counter, a fridge and trays). The sofa's back stands 0.075 m from the colonnade walkway's edge line: the walkway keeps its 1.575 m but the sofa's back is touched by everyone passing.
Evidence: `studies/courtyard_layout.typ` (the sofa at x 5.95–6.85); `3d/character.yaml` courtyard_layout.furniture.
Rule / practice: R-23 / R-24; practice.
Fix and effect: A cushion box under the hospitality kitchen's counter or a bench-box at the courtyard's south strip; move the sofa 0.2 m east (the table and the chairs follow; the 0.8 m in front of the grille becomes 0.6, the minimum). Effect: none.

V-18 · minor · **Documentation · stale records and labels**
What: `studies/stair_flip.typ` and `studies/south_wing.typ` still describe the family bath, the stair tower and the sunken seating as the state of the house (they are kept as records, the headers say so); `3d/character.yaml` views still name "the sunken seating"; the decisions log labels the owner's 3 Nov decisions V8.3 (correct) but its V8.1 rows mix the tower removal (V8.1) with items applied in V8.2 (relabelled). A reader opening a study PDF cannot always tell record from current state.
Evidence: the files named; `data/decisions.csv` rows 50–57.
Rule / practice: Practice: document control.
Fix and effect: Stamp every record study with a "RECORD · superseded by …" banner on page 1 (the stair flip, the south wing, the GF shower, bedroom 1's bath) at V9.0. Effect: none.

V-19 · suggestion · **Daylight and acoustics · no simulation yet**
What: No daylight simulation exists for the courtyard, the double-height living and the north rooms (open item 15); the living's acoustics (a 6 m high room with stone and glass) are deferred to the detail stage (E-07). Both decide finishes and the facade openings that R-37 now wants to enlarge.
Evidence: `OPEN_ITEMS.md` 15; review E-07.
Rule / practice: BS EN 17037 (daylight); practice (reverberation ≤ 0.8 s in a living room).
Fix and effect: Run a simple daylight model (the 3D exists) before the facade stage; brief an acoustician for the living's ceiling and the clerestory wall. Effect: the facade stage.

V-20 · suggestion · **Lift · supplier data missing**
What: The lift shaft 2.25 × 2.2 is a placeholder; the headroom limit (≤ 3.40 m above the first-floor stop, 3.70 with a raised head) and the pit depth are assumed; the pit meets the soil report (V-02).
Evidence: `OPEN_ITEMS.md` 14.

Rule / practice: SBC 1101 / EN 81-20 (**not verified**).
Fix and effect: Choose two suppliers now and take the larger shaft. Effect: the lobby's width if the shaft grows east (the coat cupboard).

8 · Prioritised action list

Before freezing the layout for V9.0

1. Decide bedroom 1's shower (V-03) and the suite bath's approach (V-10); redraw the first floor accordingly.
2. Decide the water system (a roof tank, the heaters, the batteries' room; V-05, V-10) and the parking bay / entrance canopy (V-07, R-11); fix the hood / outdoor counter conflict (V-08).
3. Add heating to the MEP brief and a first envelope specification (V-04); add the door at the living's arch and the clerestory actuators to the fire strategy (V-09).
4. Stamp the record studies and relabel the decisions log (V-18); move the sofa and store the cushions (V-17); a drying line (V-16).

What the engineers must do (in parallel, starting now)

1. The legal survey and the septic / tanks / levels survey (V-01, V-06).
2. The soil investigation and the first structural sizing round (V-02: X5, the projection, the trimmers, X21, the depressed zone, the roof tank and PV loads, the lift pit).
3. The MEP concept: the HRV and ducts, the plumbing design, heating, the SBC 602 path, the electrical rooms (V-04, V-05, V-08, V-10); the lift supplier (V-20).
4. The municipality and the Civil Defense: the parking, the setback, the septic, the gas, the checklist.

What can wait for the detail stage

1. The facades under R-37 (with the daylight and acoustic studies, V-19), the bathroom interiors, the kitchen joinery, the courtyard furniture, the thresholds and the entrance ramp, the guest WC and maid bath door widths (V-11), the family link (V-12), the stair trimmer depth (V-13 with the engineer), the roof ladder's hoist (V-14), the second guest WC (V-15).

قائمة العمل: قبل تجميد المخطط لـ V9.0: دش غرفة النوم 1، أسلوب حمام الجناح، نظام الماء والبطاريات، الموقف والمظلة، الشفاط والمطبخ الخارجي، التدفئة والغلاف، باب قوس الصالة ومحركات النوافذ العلوية، ختم الدراسات القديمة. للمهندسين الآن: الرفع المساحي، التربة والإنشاء، التهوية والسباكة والتدفئة والكهرباء، مورد المصعد، البلدية والدفاع المدني. لمرحلة التفاصيل: الواجهات والحمامات والمطبخ والأثاث والعتبات.

Interim verification and validation review of V8.3 · 3 Nov 2026 · review only, nothing here changes a design file · concept design, not for construction; every structural and MEP size in the project is assumed until the engineers of record confirm it. The next step is V9.0 after the "before freezing" items, then the full V&V review.